

## Question

Calcium carbonate is also one of the important components of scale. A large amount of scale may cause uneven heat transfer, and even cause explosions in boilers. Therefore, efficient water softening is one of the important topics.

Given the following thermodynamic data: (Air  $p(\text{CO}_2) = 0.034 \text{ kPa}$ )

| Substance                          | $\Delta_f H_m^\ominus / \text{kJ} \cdot \text{mol}^{-1}$ | $S_m^\ominus / \text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$ |
|------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------|
| $\text{CaCO}_3(\text{s})$          | -1207.6                                                  | 91.7                                                               |
| $\text{Mg}(\text{OH})_2(\text{s})$ | -924.7                                                   | 63.24                                                              |
| $\text{H}_2\text{O}(\text{l})$     | -285.8                                                   | 69.95                                                              |
| $\text{CO}_2(\text{g})$            | -393.5                                                   | 213.8                                                              |
| $\text{Ca}^{2+}(\text{aq.})$       | -543.0                                                   | -56.2                                                              |
| $\text{Mg}^{2+}(\text{aq.})$       | -467.0                                                   | -137.4                                                             |
| $\text{OH}^-(\text{aq.})$          | -230.0                                                   | -10.90                                                             |
| $\text{CO}_3^{2-}(\text{aq.})$     | -675.2                                                   | -50.0                                                              |
| $\text{HCO}_3^-(\text{aq.})$       | -689.9                                                   | 98.4                                                               |

Hard water often leaves scale in kettles. If boiling ( $T = 373 \text{ K}$ ) water with  $\text{pH} = 7.35$  at a certain normal temperature ( $T = 298 \text{ K}$ ) (the water only contains carbonate buffer pairs) does not produce  $\text{CaCO}_3$  or  $\text{Mg}(\text{OH})_2$  scale, find the maximum hardness of the water (here it refers to the sum of the concentrations of calcium and magnesium ions in the water, which needs to be converted into the mass of calcium carbonate, in units of ppm, i.e.,  $\text{mg/L}$ ).

A. 730 ppm

B. 430 ppm

C. 7.3 ppm

D. 300 ppm

E. 350 ppm

F. 600 ppm

G. 3.5 ppm

## Answer

Correct Answer: **A**

## Detailed Explanation

First, calculate the equilibrium constant of the reaction:  $CO_2(g) + OH^-(a.q.) = HCO_3^-(a.q.)$

Reaction  $\Delta_r H_m^\ominus = -66.4 \text{ kJ} \cdot \text{mol}^{-1}$ ,  $\Delta_r S_m^\ominus = -104.5 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$

### CHECKPOINT

1 PTS

$$\Delta_r H_m^\ominus = -66.4 \text{ kJ} \cdot \text{mol}^{-1}, \quad \Delta_r S_m^\ominus = -104.5 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$$

At 298 K, the reaction has  $\Delta_r G_m^\ominus = -35.3 \text{ kJ} \cdot \text{mol}^{-1}$ ,  $K_1^\ominus = 1.5 \times 10^6$

### CHECKPOINT

1 PTS

$$\text{At } 298 \text{ K, } \Delta_r G_m^\ominus = -35.3 \text{ kJ} \cdot \text{mol}^{-1}, \quad K_1^\ominus = 1.5 \times 10^6$$

At 373 K,  $\Delta_r G_m^\ominus = -27.4 \text{ kJ} \cdot \text{mol}^{-1}$ ,  $K_2^\ominus = 6.9 \times 10^3$

### CHECKPOINT

1 PTS

$$\text{At } 373 \text{ K, } \Delta_r G_m^\ominus = -27.4 \text{ kJ} \cdot \text{mol}^{-1}, \quad K_2^\ominus = 6.9 \times 10^3$$

It is easy to calculate that the solution at room temperature contains  $[HCO_3^-]_1 = 1.14 \times 10^{-4} \text{ mol} \cdot \text{L}^{-1}$

#### CHECKPOINT

1 PTS

The solution at room temperature contains  $[HCO_3^-]_1 = 1.14 \times 10^{-4} \text{ mol} \cdot \text{L}^{-1}$

After boiling, the equilibrium shifts,  $[HCO_3^-]_2 + [OH^-]_2 = [HCO_3^-]_1 + [OH^-]$

Solving for  $[HCO_3^-]_2 = 8.0 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1}$   $[OH^-]_2 = 3.4 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1}$

#### CHECKPOINT

1 PTS

After boiling,  $[HCO_3^-]_2 = 8.0 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1}$   $[OH^-]_2 = 3.4 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1}$

Consider the reaction  $HCO_3^-(a.q.) + OH^-(a.q.) = CO_3^{2-}(a.q.) + H_2O(l)$

The equilibrium constant of the reaction at 373 K is  $K^\ominus = 1.7 \times 10^2$ , at this time  $[CO_3^{2-}] = 4.6 \times 10^{-7} \text{ mol} \cdot \text{L}^{-1}$

#### CHECKPOINT

1 PTS

At 373 K,  $K^\ominus = 1.7 \times 10^2$ , at this time  $[CO_3^{2-}] = 4.6 \times 10^{-7} \text{ mol} \cdot \text{L}^{-1}$

Then consider  $Ca^{2+}(a.q.) + CO_3^{2-}(a.q.) = CaCO_3(s)$

The equilibrium constant of the reaction at 373 K is  $K^\ominus = 7.1 \times 10^8$ , under critical conditions  $[Ca^{2+}]_{\text{max}} = 3.0 \times 10^{-3} \text{ mol} \cdot \text{L}^{-1}$

Consider the reaction  $Mg^{2+}(a.q.) + 2OH^{-}(a.q.) = Mg(OH)_2(s)$

The equilibrium constant of the reaction at 373 K is  $K^{\ominus} = 2.0 \times 10^{11}$ . Under critical conditions,  $[Mg^{2+}]_{\max} = 4.3 \times 10^{-3} \text{ mol} \cdot \text{L}^{-1}$

CHECKPOINT

1 PTS

Under critical conditions,  $[Mg^{2+}]_{\max} = 4.3 \times 10^{-3} \text{ mol} \cdot \text{L}^{-1}$

Adding the calculated critical calcium and magnesium ion concentrations, the maximum hardness of water is = 730 ppm

CHECKPOINT

1 PTS

The maximum hardness of water = 730 ppm